TAMID QUANT — STRATEGY ONE PAGER

Bitcoin Digital Options: Selling the Kalshi Longshot Tails

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The Idea

Kalshi's hourly "Bitcoin above $K at 5pm ET" contracts overprice longshots in the last hours before expiry. Contracts quoted at 1.5-2.5c settled YES 0.14% of the time (1 of 698), and the low tail 3h out priced 1.7% against 0.4% realized; 12h out the tails are fairly priced. We sell overpriced low-tail YES and buy underpriced high-tail YES, and size each day's book so the worst possible BTC settlement loses at most 3% of bankroll. Over 50 live days (Aug 26 to Oct 14, 2026) it made +2.4% with no losing day, but no tail event occurred, so the risk side is untested (see Key Risks).

Mathematical Framework

Core Model: *A contract pays $1[S_T > K]$. For positions q_i with side s_i (+1 buy, −1 sell YES) at fill p_i, the book's payoff if BTC settles in scenario j is*

$\Pi_j = \Sigma_i\, q_i \cdot [s_i \cdot (1[S_j > K_i] - p_i)]$ − fees. Only m+1 scenarios matter (one between each pair of the day's m strikes), so the worst case $\min_j \Pi_j$ is exact, not simulated.

Signal Construction: *Walk-forward tail probability per (horizon, price bin), using prior expiry days only:*

$f = (\Sigma y + N_0 \cdot m) / (n + N_0)$, $N_0 = 50$, where m is the bin's average market price. It starts at the market price (no edge) and learns from outcomes. Edge = f vs. the sizing price (quote ± 2c slippage, clipped to 1c/99c), less fee.

Entry / Exit Logic: *Decide 6h and 3h before expiry, fill one hour later at the Kalshi bid or ask:*

Trade contracts with mid < 10c or > 90c, spread ≤ 6c, edge ≥ 0.5c and at least 100 prior observations in the bin, after a 30-day warm-up. Hold to settlement.

Position Sizing: *Linear program per expiry day:*

maximize $\Sigma\, q_i \cdot edge_i$ subject to $base_j + \Sigma\, q_i \cdot \Pi_{ij} \geq -3\%$ of bankroll in every scenario j, $0 \leq q_i \leq 50$. Positions already held from the 6h round enter as base. Nested strikes share one budget; a two-sided book (rally and crash risk) gets more room. Whole-contract rounding is re-checked, and 1c of cap is reserved per order for fee rounding.

Key Assumptions

- Tail mispricing seen on earlier days persists; the estimator learns only from past expiries
- Hourly-candle quotes are fillable at 1c / 99c for about 30 contracts (no depth data); fee = $0.07 \cdot q \cdot P \cdot (1 - P)$, rounded up to the cent per order
- Each expiry day is one draw, so the effective sample is 50 live days, not 73 trades

Key Risks

- No tail event in the sample: every sell settled NO and every buy settled YES, so the 0% drawdown says nothing. The 3% cap is the real protection
- About +1.1c per contract won against about −99c lost: one worst-case day (−2.97%) erases about 62 average days (+0.048%/day). Breakeven is roughly 1.1% of contracts going against us; at the 1c price floor the market was already calibrated (0.63% realized)
- The edge is partly a shrinkage artifact: f falls as tail-free days accumulate. All 73 trades sit at the 1c/99c clip, so the 2c slippage buffer never applied
- Per-trade sizing earns more (+7.5%) but on 30 days a single BTC move would have cost up to 29.5% of bankroll, because nested strikes lose together

At a Glance

Asset Class:	*Crypto / Prediction Markets*
Strategy Type:	*Longshot Bias / Tail Risk Premium*
Direction:	*Short Low-Tail YES, Long High-Tail YES*
Holding Period:	*3-6 Hours (Hold to Expiry)*
Universe:	*Kalshi KXBTCD Tail Strikes*

Backtest Results (50 live days)

Sizing	Return	Max worst case	Days > 3%
Portfolio cap	+2.4%	2.97%	0
1 contract / signal	+0.1%	1.0%	0
Per-trade 3% cap	+7.5%	29.5%	30

$1,000 bankroll; 3% cap = $30 worst case per expiry day.

Data Requirements

- Kalshi KXBTCD hourly bid, ask and settlement - public API
- 120 expiries (Jun to Oct 2026), 6,242 quotes at 6h and 3h; the first ~70 days only train the estimator
- Order-book depth (not yet collected)

Key Parameters (starting values)

- Worst-case cap: 3% of bankroll per expiry day
- Tails: mid < 10c or > 90c; 6h and 3h rounds
- Shrinkage $N_0 = 50$; min 100 obs per bin; 30-day warm-up
- Min edge 0.5c; slippage 2c; max 50 contracts per strike per round

Open Questions

- How often do 1c tails hit over a longer history (2+ years)?
- Is there depth to sell 30+ contracts at 1c?
- Can resting (maker) orders beat the 1c / 99c fills and fees?
- Does the same bias exist in Kalshi's hourly S&P 500 contracts (KXINXU)?